B.Sc. (Hons). Computer Science

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ElderConnect

Contextual Report

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Abstract

The aim of this project is to develop a mobile application named ElderConnect, which is designed specifically for targeting elderly users who are aged 60 and above. This application offers an all-in-one digital platform where the elderly can connect with their friends, access personalized news and manage their health. Many elderly people experience loneliness, difficulties using modern applications, and challenges in maintaining their regular health routines. This proposed system would provide a friendly and interactive environment that would help users to be socially connected, mentally active, and health conscious.

The application features a colour-coded interface with large icons and guidance for ease of use. It includes two main interfaces, one for elderly users and another for guardians or caretakers, which is selectable during registration. Elderly users have the ability to choose what interests them be it news, hobbies or sport, which the Artificial Intelligence recommendation model then utilizes to tailor the feed. It has an application that is a social feed, a customized news section and a text to speech option that is used by the visually impaired. The mood detection model is an AI-based model that analyzes user posts and messages in order to discover moods such as happy, sad, or neutral. This information is displayed on the caretaker dashboard with permission to enable them to track the emotional health of the user. The app will also have a health and medication reminder feature and a wellness section that comprises basic memory games, breathers, and relaxers.

Keywords: Elderly Care, Mobile Application, Artificial Intelligence, Sentiment Analysis, Social Isolation, Medication Reminder, Accessibility

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Introduction

1.1 Background

The demographic transition of the world population is also facing an ageing shift. The World Health Organization (2022) also estimates that the number of persons aged 60 and above will grow to 2.1 billion by 2050, almost twice the number of 1 billion in 2020. This expansion of the elderly population results several challenges particularly as, social isolation, mental health deterioration, and the management of daily health routines. A considerable number of older adults now live independently or at significant distances from their families due to workforce mobility and modern lifestyle changes.

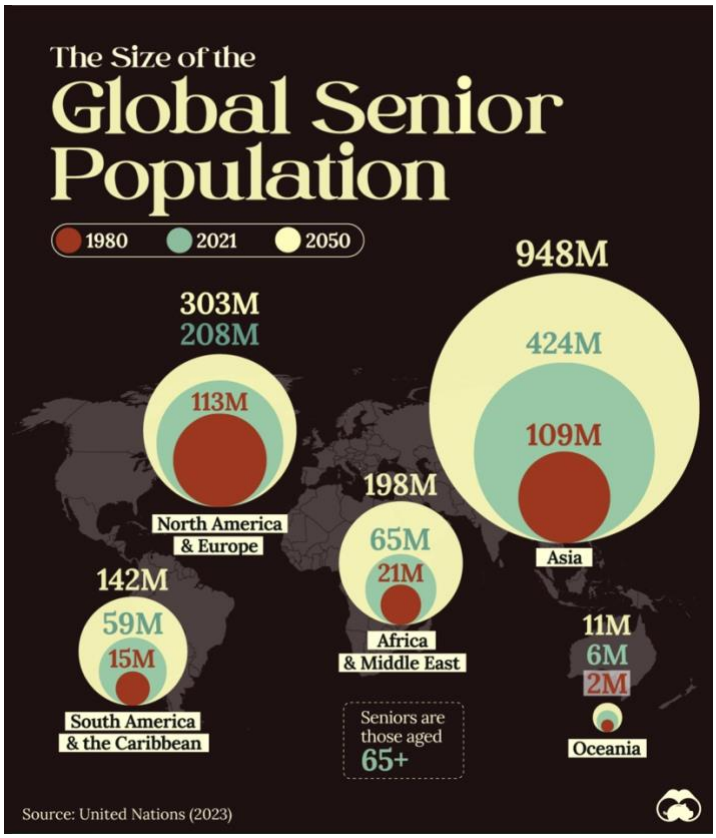

Figure 1: Global Senior Population

Loneliness has become a serious health concern for elderly individuals. Sen, Prybutok and Prybutok (2021) found that social isolation can badly affect both the physical and mental health of older adults. It can lead to depression, memory problems, and other health issues. because

of the complicated designs and confusing interfaces of many digital tools available today for communication, most elderly people find them hard to use even though there are many varieties

Technology has the potential to solve these issues and smartphones are becoming more common among older people. The research shows that more elderly users are trying to adopt them (Chua, Lim, and Teh, 2023). However, popular applications as Facebook, WhatsApp, and Instagram are not designed targeting for this age group. These applications often feature small typography, cluttered navigation, and overwhelming notification systems that can deter the users with limited digital literacy.

Another issue is that most existing apps for elderly care only focus on one thing. Some applications exclusively focus on social networking, while others address only medication adherence. There is no single platform that brings together social connection, emotional support, and health management in a way that is genuinely easy for elderly users to handle.

This project, 'ElderConnect', is aiming to solve these problems by creating an AI-powered mobile application made specifically for people aged 60 and above. By combining social interaction, mental wellness activities, and health management tools in one place along with AI–based.

1.2 Motivation

The motivation for this project is derived from both personal observation and academic research findings. In many communities, elderly individuals experience loneliness and social disconnection particularly when living alone. While smartphones are becoming more accessible to older adults, the complexity of existing applications prevents many from taking full advantage of these devices.

Existing solutions in the market consists of several limitations. Applications like GrandPad require specialized devices that may not be affordable for all users. Platforms such as Connect2Affect by AARP focus primarily on social interaction without addressing mental wellness or health management. While general social media platforms such as Facebook are too complex and cluttered for elderly users to navigate comfortably (Care 365 Homecare, 2025).

Accordingly, the emotional well-being of the elderly users is often overlooked in existing applications. Many of them experience the feelings of sadness or loneliness but may not tend to communicate those emotions openly to their family. So, an AI-powered system capable of detecting emotional patterns from the user's behavior (text based) could help to identify when the user might need additional support which allowing caretakers to respond appropriately.

These factors conclude a clear need for a unified, accessible platform that addresses social, emotional, and health needs simultaneously. This proposed project aims to deliver such a solution.

1.3 Problem Statement

Even though there are plenty of social media platforms and health applications out there, elderly users still face major challenges when trying to use digital technology. The core problem this project addresses can be summarised as follow,

How a mobile application can be designed to effectively support social engagement, emotional well-being, and health management for elderly communities while remaining simple and accessible enough to use independently?

This problem directs to several interconnected challenges:

1. **Usability Barriers -** Existing platforms feature small text, complicated menus, and overwhelming interfaces that are difficult for elderly users to navigate through.
2. **Lack of Unified Solutions -** Current applications address only per one aspect of elderly care, requiring users to install and learn multiple applications for different purposes.
3. **Emotional Oversight -** Most existing platforms do not monitor or support the emotional health of elderly users, results missing the opportunities to identify signs of loneliness or distress.
4. **Limited Caretaker Involvement -** Lack of simple tools to monitor and support the elderly relatives by their Family members and caretakers remotely without being intrusive.
5. **Accessibility Gaps:** Rarely integrated features such as text-to-speech for visually impaired users and voice input options in mainstream applications.

Addressing these challenges requires a solution designed from the ground up with elderly accessibility as the core principle, rather than adapting existing platforms that were created for younger demographics.

1.4 Aim

The aim of this project is to develop an AI-based social and wellness mobile platform that promotes social interaction, emotional well-being, and health awareness among elderly users through a simple, accessible, and personalised interface.

1.5 Objectives

To achieve this aim, the following objectives have been defined:

1. To design and implement an intuitive and accessible mobile interface tailored for elderly users, featuring large buttons, colour-coded navigation, and clear icons.
2. To develop an AI-based sentiment analysis module for mood detection by analysing text content such as posts and messages.
3. To personalise news and content recommendations based on user interests selected during registration.
4. To integrate a text-to-speech feature to support visually impaired users in consuming content.
5. To include a wellness and mindfulness section featuring memory games, breathing exercises, and relaxation activities.
6. To implement a medication reminder system with visual and voice alerts connected to a caretaker interface.
7. To develop a caretaker dashboard enabling family members or guardians to track user mood, medication adherence, and daily activity.

1.6 Description of the Artefact

The proposed artefact is a cross-platform mobile application built using the Flutter framework, with MongoDB as the backend database and MongoDB Atlas App Services for secure user authentication and session management.

The application consists of two main interfaces designed to serve different user groups as,

1.6.1 Elderly User Portal

This is the main interface, which has been designed keeping simplicity as the main principle. Featuring,

- **Simplified Registration:** A color-coded process of sign up with visual and voice instructions, where the users can choose their interests like news, hobbies, or sports.
- **Social Feed -** Users can share posts, photos, and greetings with friends and family in a clean, easy-to-read format.
- **Personalized Content -** Tailored news and entertainment content based on user interests using AI-based recommendations.
- **Text-to-Speech -** Visually impaired users can read aloud the content.
- **Talk Button -** Allows sending voice messages (stored as audio only) to replace typing efforts. These recordings are not processed by the AI sentiment analysis module.
- **Wellness Section -** Memory games, breathing exercises, and relaxation activities to promote mental wellness.
- **Medication Reminders -** Visual and voice notifications to remind users when to take their medications.
- **Emergency Contact -** A prominent button for quickly contacting family members or caretakers in case of emergency.

- **AI Mood Detection -** The system analyzes texts based on the posts to identify emotional patterns, with consent, sharing mood summaries with caretakers.

1.6.2 Caretaker Portal

This secondary interface allows family members or caretakers to support their elderly relatives remotely. Features include,

- **Medication Management:** Setting up medication schedules including pill names, colours, dosages, and timings.
- **Mood Monitoring:** Viewing emotional status summaries based on AI analysis (with user consent).
- **Activity Summary:** Tracking general app usage and engagement levels.

1.7 Scope

1.7.1 In Scope

The following items are included within the scope of this project:

- Development of a fully functional mobile application for Android and iOS platforms using Flutter
- Social feed functionality for sharing posts, photos, and messages
- AI-powered sentiment analysis for mood detection from text
- Content personalisation based on user preference
- Text-to-speech functionality for accessibility
- Voice message feature for easier communication (audio storage only, no AI processing)
- Wellness section with memory games and relaxation exercises
- Medication reminder system with visual and voice notifications
- Caretaker dashboard with mood and activity monitoring features
- User authentication and storage with MongoDB

1.7.2 Out of Scope

The following items are outside the scope of this project:

- Voice-based sentiment analysis (only text-based analysis is included, voice messages are stored as audio files without AI processing)
- Integration with medical devices or wearable technology
- Real-time video calling features
- Multi-language support (English only for this version)
- Integration with external healthcare systems or electronic health records

1.8 Proposed Solution

To address the identified problems, this project proposes an all-in-one mobile application that combines multiple features elderly users need in their daily. The solution is designed based on the following principles.

Accessibility First - Every design priorities ease of use for elderly individuals, including larger buttons, high contrast colors, simple navigation paths, minimal text, and voice support features.

AI-Powered Personalisation - The application uses Artificial Intelligence to personalise the content feed based on user interests and detects emotional patterns from text. This makes the experience more supportive for the user.

Connected Care - The caretaker dashboard creates a bridge between elderly users and their family members, enabling remote support and monitoring without being intrusive or violating user's privacy.

Holistic Wellness – Instead of focusing on a single aspect of elderly care, the application addresses social, emotional, and health needs together in one platform.

Privacy and Consent - All sensitive features, particularly mood sharing with caretakers, operates on a consent-based model where users have full control over what information is shared.

1.9 Structure of the Report

This contextual report is organized as into the following sections:

Section 1 – Introduction - Provides an overview of the project including background study , motivation, problem statement, aim, objectives, description of the artefact, scope, and proposed solution.

Section 2 - Literature Review and Market Research - Presents a comprehensive review of existing academic research related to elderly technology use, AI in healthcare applications, accessibility design principles, and similar existing systems. This section also includes primary research findings from surveys conducted with elderly users and caretakers.

Section 3 - Project Plan - Outlines the project timeline, tasks, and milestones using a Gantt chart and Work Breakdown Structure (WBS).

Section 4 - Planning the Artefact - Describes the development methodology, technical architecture, tools and technologies, design considerations, implementation approach, testing strategy, and risk mitigation techniques.

References - Lists all sources cited in this report using Harvard referencing style.

Appendices - Contains supplementary materials including survey questionnaires and system diagrams.

Literature Review and Market Research

This section presents a comprehensive review of existing academic researches and market analysis which relevant to the ElderConnect project. The literature review examines the main challenges the ageing population faces and how elderly users interact with digital technology. It also covers the principles of accessible mobile design, the role of Artificial Intelligence in healthcare, and an analysis of similar systems currently in the market. Following this review, the section presents findings from surveys conducted with elderly users and caretakers to confirm the real-world need for the proposed solution.

The purpose of this section is to establish a strong theoretical foundation for the project while identifying gaps in current solutions that ElderConnect aims to address.

2.1 Literature Review

2.1.1 Introduction

The literature review is structured into several research areas that are directly relevant to the development of ElderConnect. These areas include understanding the ageing population and their challenges, exploring how elderly users interact with technology, examining design principles for the accessible applications, investigating the use of AI for health monitoring, and analyzing existing systems in the market.

The following subsections cover:

- Research Area A: Ageing Population and Social Isolation
- Research Area B: Elderly Users and Technology Adoption
- Research Area C: Accessible Interface Design for Elderly
- Research Area D: Artificial Intelligence in Healthcare
- Research Area E: Sentiment Analysis and Mood Detection
- Research Area F: Similar Existing Systems

2.1.2 Research Area A: Ageing Population and Social Isolation

The Growing Elderly Population

The world is currently going through a major shift towards an older society. According to the United Nations (2019), the number of people who are aged 65 and over is expected to be more than double in amount by 2050, reaching over 1.5 billion worldwide. This demographic transition is occurring in almost every country, while some nations like Japan and those in Europe are already experiencing a substantial elderly population.

While increased life expectancy is generally considered as a positive fact, it also indicates improvements in healthcare and living conditions. However, living longer also means more years dealing with age-related health problems, reduced mobility, and often increased isolations from others (Beard et al., 2016).

Understanding Social Isolation and Loneliness

Social isolation and loneliness are related but two distinct concepts. Social isolation is an objective term to the lack of contact with other people, whereas loneliness is the subjective term which is the feeling of being alone even while surrounded by others (Holt-Lunstad et al., 2015).

These conditions are especially common among elderly people which seriously effects on their health and well-being. Holt-Lunstad et al. (2015) conducted a comprehensive meta-analysis by examining data from many studies which shows loneliness increases the likelihood of death by 26%, and social isolation increases it by 29%. These numbers showcase the seriousness of this issue comparable to the risks those are associated with smoking or obesity,

Sen, Prybutok and Prybutok (2021) conducted a systematic review about using digital technology for the social well-being of adults. Their research found that social isolation elevates the rates of depression, anxiety, and cognitive decline. They also found that if the digital tools are designed appropriately for elderly users, which can effectively reduce feelings of loneliness.

Factors Contributing to Social Isolation in Elderly

- Older people often experience the loss of their spouses, siblings, and close friends which cause to naturally shrink their social circle and the available support network over time.

- Health problems and reduced mobility make elderly people difficult to leave their homes and engage in community activities or maintain social connections.

- Geographic separation is another significant issue as many elderly people live at significant distances from their children and relatives due to migration for work or other circumstances.

- Retirement results in the loss of daily interactions with colleagues that many people previously took for granted.

- While technology offers new ways to stay connected, majority of elderly users find modern applications too confusing to use in daily basis (Chua, Lim and Teh, 2023).

2.1.3 Research Area B: Elderly Users and Technology Adoption

Current State of Technology Adoption

Despite common assumptions that states elderly people are resistant to technology, researches indicates that smartphone adoption among older adults are steadily increasing. Chua, Lim and Teh (2023) found that smartphone ownership among people aged 65 and above has grown considerably in recent years.

However, simply owning a device does not say it is being used effectively. A considerable number of elderly smartphone owners limit themselves to basic functions like making calls and rare interaction with applications.

Anderson and Perrin (2017) found that although 42% of adults over 65 those who owned smartphones, only a small percentage use confidently for the tasks beyond basic communications. This gap between ownership and effective usage shows a significant opportunity for applications designed specifically to this age group.

Barriers to Technology Adoption

Several types of barriers which prevent elderly users from fully adopting to digital technologies.

Physical Barriers:
Many elderly individuals have vision problems, resulting difficulties of reading small texts and tiny icons. Some have reduced motor control, making it challenging to tap small buttons accurately. Others experience hearing difficulties, resulting in missed audio notifications.

Cognitive Barriers:
Memory issues make it difficult to remember application navigation or passwords. Slower processing speed means complicated interfaces can feel overwhelming. Additionally, many elderly users experience anxiety about pressing the wrong button and causing problems.

Design-Related Barriers:
Most applications today are designed for younger demographics. Interfaces are often

cluttered with numerous options. Clear instructions are rarely built into applications. Different applications work differently, creating inconsistency that confuses users.

Yang, Yang and Li (2025) analyzed elderly behavior and satisfaction with adapted mobile applications to them. Their research found that elderly users showed significantly higher satisfaction rates for the applications designed specifically for them. A considerable difference has made using features such as simple navigation, larger buttons, and clear feedback.

The Digital Divide

"Digital divide" is the term that describes the gap between those who can use digital technology effectively and those who cannot. This issue is especially critical for elderly populations, as more everyday services are moving online that are leaving those without digital skills at risk of being left behind.

A randomized controlled trial by Kawaguchi et al. (2024) conducted to test the effects of a mobile application designed to promote social participation among older adults. As a result, The increase of social engagement and reduction of loneliness by the well-designed mobile applications successfully. However, the study also highlights that these benefits depends significantly on the usability of the application for the target age group.

2.1.4 Research Area C: Accessible Interface Design for Elderly

Principles of Accessible Design

Applications designed for elderly users requires a careful attention to the accessibility principles. The World Wide Web Consortium (W3C) provides the Web Content Accessibility Guidelines (WCAG) that can be applied when mobile application development is done. These guidelines emphasize the four key principles as, perceivable, operatable, understandable, and robust (W3C, 2018).

Certain design elements are particularly important for elderly users specifically.

Visual Design Considerations:
Large texts should be used at minimum 16 points while maintaining high contrast between text and background to improve readability. simple and easily recognizable Icons should be used. Furthermore, colour alone should not be the only method to convey information, as some users may have colour vision deficiencies.

Interaction Design Considerations:
Buttons and touch targets should be large, according to Apple's guidelines at least 44x44 pixels. Navigation should be simple without excessive hierarchy. When users perform an action, clear feedback should indicate the result. Complicated gestures requiring multiple fingers should be avoided.

Content Design Considerations:
Language should be simple and easy to understand while the sentences should be concise. Headings should be clear and descriptive. Layout should remain consistent across different screens to prevent confusion.

Research on Elderly-Friendly Interface Design

Chua, Lim and Teh (2023) specifically investigated the design of age-friendly mobile applications which found valuable insights of what makes an application suitable for elderly users.

Their findings indicate that elderly users prefer applications that perform fewer functions but execute them effectively. They do not want applications with numerous features they cannot understand while consistency is also important; If a button appears a certain way on one screen, it should appear the same way on every screen. This helps users build confidence.

Built-in guidance and clear instructions significantly improve user experience. While many applications assume that the users already understand how to navigate them, but elderly users often require step-by-step guidance. Furthermore, designing carefully is more effective to prevent mistakes than allowing users to recover from errors.

Moreover, for their personal preference it is crucial to allow elderly users to adjust settings, like text size and contrast.

Gaver et al. (2013) discussed the importance of human-centered design approaches when creating products for specific user groups. when developers designing for a particular demographic, they must understand their actual needs rather than making assumptions which is the key principle. This approach is especially relevant for elderly users, those who needs and often misunderstood navigation.

Colour-Coded Navigation

One design approach that has demonstrated effectiveness for elderly users is colour-coded navigation. Assigning different colours to different sections of an application. For an example, the social section might be blue, the health section might be green. As a result, the users can remember associations like "blue is for chatting with friends" even if they forget the exact button location.

Hawthorn (2000) researched this approach and found that colour coding helps elderly users navigate faster and make fewer mistakes. However, colours must be chosen carefully. Some elderly individuals have difficulty distinguishing certain colours, so colour should never be the only method of differentiation. Text labels and icons should also be included.

2.1.5 Research Area D: Artificial Intelligence in Healthcare

Overview of AI in Healthcare

Artificial Intelligence is being applied increasingly in healthcare field for various purposes including diagnosis, treatment recommendations, patient monitoring, and administrative tasks. According to Topol (2019), to transform healthcare, AI has a significant potential by enabling more accurate diagnoses, personalized treatments, and improved patient outcomes.

AI offers several promising applications, specifically for elderly care,

Analyzing and monitoring user's health through various data sources to detect changes and predict potential problems before they become serious. Medication management by sending reminders and tracking adherence. Emotional support by detecting mood patterns of the user and providing appropriate responses and learning user preferences and behaviors to provide tailored content and recommendations.

AI Applications for Elderly Well-being

The application of AI specifically for elderly well-being is a growing field of research. Studies have explored various approaches including conversational agents, activity monitoring, and sentiment analysis.

Vardoulakis et al. (2012) developed a conversational agent designed to provide companionship for an isolated older group. The system engaged with users in casual conversations and able to detect emotional cues in their responses. While at that time the technology had limitations, and the study demonstrated the potential for AI-based systems to provide social and emotional support.

Recent advances in natural language processing and machine learning have made such applications considerably more feasible. Pre-trained language models and sentiment analysis APIs are now allowing the developers to incorporate sophisticated AI capabilities into applications without need to build everything from scratch.

2.2.6 Research Area E: Sentiment Analysis and Mood Detection

Understanding Sentiment Analysis

Sentiment analysis, which also known as opinion mining, which is a natural language processing technique that used to identify and extract subjective information from texts. At its most basic level, this analysis classifies texts as positive, negative, or neutral while more advanced systems can detect specific emotions such as happiness, sadness, anger, fear, and surprise (Liu, 2012).

This process typically involves several steps. The text is cleaned first and prepared for analysis. The system then identifies relevant features in the text that indicate sentiment values, such as

specific words, phrases, or patterns. Finally, a machine learning model or a rule-based system classifies the sentiment which based on the extracted features.

Applications in Health and Well-being

Sentiment analysis has been applied in various health-related contexts. Researchers have used it to analyse social media posts for signs of depression, monitor patient satisfaction, and detect emotional distress.

Calvo et al. (2017) reviewed the use of natural language processing for mental health applications. The review found that sentiment analysis shows promise for detecting emotional states that may indicate mental health concerns. However, the researcher also noted several challenges including the need for context-specific training data and also noted the importance of considering cultural and individual differences in emotional expression.

For the ElderConnect application, sentiment analysis will be used to analyze text content created by users, such as posts and messages. The system identifies patterns that suggest negative emotions like sadness or loneliness, which can provide subtle feedbacks to users with their consent, notify caretakers who may be able to provide support.

Available Tools and APIs

Several tools and APIs are available for implementing sentiment analysis in applications:

Hugging Face provides access to pre-trained natural language processing models including sentiment analysis models. The platform offers both free and paid options (Hugging Face, 2025).

Google Cloud Natural Language API offers sentiment analysis as part of a broader natural language processing service. The API can analyze text and return sentiment scores along with other information.

IBM Watson Natural Language Understanding provides another enterprise-level solution for sentiment analysis and other natural language processing tasks.

For the ElderConnect project, the Hugging Face API has been selected due to its accessibility, quality of pre-trained models, and suitability for the project scope.

2.1.7 Research Area F: Similar Existing Systems

This section analyses existing applications and systems that address aspects of elderly care, social connection, or health management. Understanding currently available solutions helps identify gaps that ElderConnect can address.

<u>General Social Media Platforms:</u>

Facebook

Facebook is the globally most widely used social networking platform, users across all the age groups. Some elderly individuals use Facebook to stay connected with family and friends however, the platform presents several challenges for that age group.

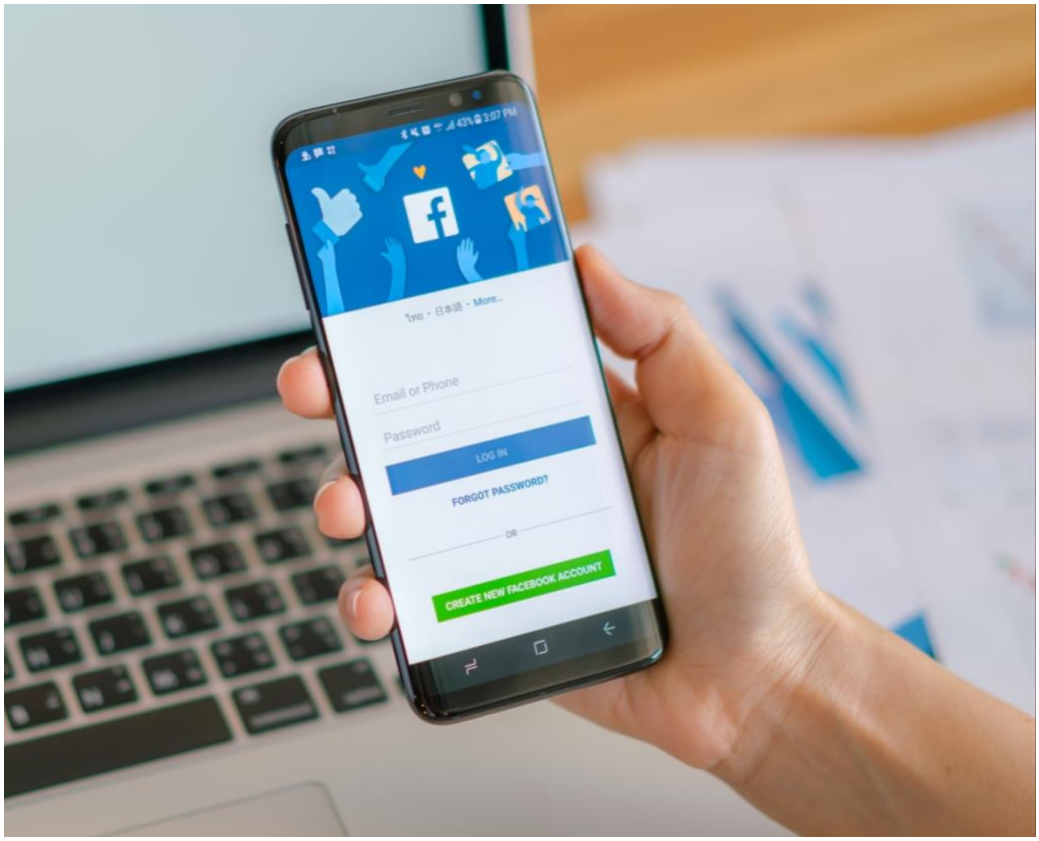

Figure 2: Interface of Facebook

The interface is complex with numerous features and options. Default text sizes are small which is difficult to use for elders with vision problems. The users who had become accustomed to previous layouts, confuse by frequent interface changes. The privacy settings make the user complicated to understand and manage while advertisements and sponsored content can be confusing or misleading.

Even though Facebook has introduced some accessibility features, the platform was not designed specifically for elderly users, and many users find it overwhelming.

WhatsApp

WhatsApp is a popular messaging application that some elderly users adopt because their family members use it. The application is relatively simpler than Facebook but still presents challenges:

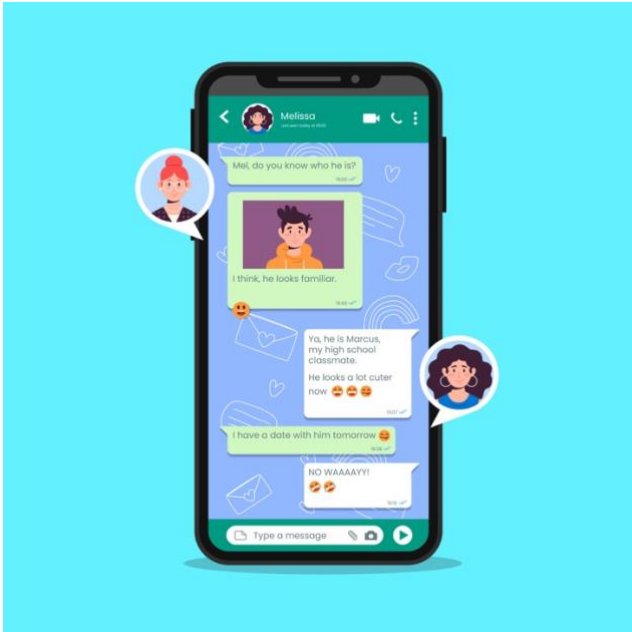

Figure 3: Interface of WhatsApp

Icons and buttons are small and difficult to tap accurately. Text input is challenging for users with typing difficulties. Group conversations can become confusing with many participants. The application lacks features for health management or mood monitoring.

<u>Elderly-Specific Applications</u>

GrandPad

GrandPad is a tablet-based service designed specifically targeting elderly community. The system features a simple, clutter-free interface with large icons, video calling with family members, photo sharing capabilities, and voice-activated features.

GrandPad

Pros

Cons

Buy now at Amazon

$399.99 from Walmart

GrandPad (**available at Amazon**) occupies a curious space as one of the only tablets specifically designed for the senior generation. While mainstream products like **Apple's iPad** or **Amazon's Fire Tablet** can certainly be retrofitted to better support older adults, this product is meant for those over the age of 75 who truly know little to nothing about technology and don't have the ability or desire to master the complexities of app stores, account passwords, or firmware updates.

Chances are you or someone you know fits this description, and **GrandPad** aims to capture that niche market with a completely tailored, subscription-based experience with just similarities to a feature-complete tablet.

$400 at Amazon

Figure 4: GrandPad product review

However, GrandPad requires purchasing a proprietary device, creating a significant cost barrier. The service also requires a monthly subscription. This model limits accessibility for elderly users who cannot afford the hardware and ongoing subscription costs.

Connect2Affect by AARP

The platform 'Connect2Affect' is an initiative by AARP (American Association of Retired Persons) that focuses on addressing social isolation among older adults. This provides resources for resolving social isolation risks, tools for finding local community activities, and provide information about the volunteer opportunities.

It is primarily a resourceful website rather than a comprehensive mobile application and It does not include features such as health management, medication reminders, or emotional well-being monitoring, even though the Connect2Affect addresses the important issues of social isolation, (Connect2Affect, 2025).

Medisafe

Medisafe is a medication reminder application that helps users to track their medication schedules. Features including of customizable medication reminders, pill identification features, adherence tracking and reports correctly, and caretaker notifications.

While Medisafe addresses medication management effectively, it does not include social features, wellness activities, or mood monitoring. Users requiring these additional features would need to use multiple applications.

Comparison of Existing Systems

The following table compares the features of existing systems with the proposed ElderConnect application:

Table 1: Existing Systems Comparison

Feature	Facebook	WhatsApp	GrandPad	Connect2Affect	Medisafe	ElderConnect
Social Networking	✓	✓	✓	Limited	✗	✓
Elderly-Friendly Design	✗	✗	✓	Partial	Partial	✓
Text-to-Speech	Limited	✗	✓	✗	✗	✓
Voice Messages	✓	✓	✓	✗	✗	✓
Medication Reminders	✗	✗	✗	✗	✓	✓
Wellness Activities	✗	✗	Limited	✗	✗	✓
AI Mood Detection	✗	✗	✗	✗	✗	✓
Caretaker Dashboard	✗	✗	Limited	✗	✓	✓
Personalized Content	✓	✗	Limited	✗	✗	✓
Works on Regular Phones	✓	✓	✗	✓	✓	✓
Free to Use	✓	✓	✗	✓	Partial	✓
Emergency Contact Button	✗	✗	✓	✗	✗	✓

As demonstrated in this comparison table, no existing solution offers everything that elderly users require. Some applications are too complicated. Some require purchasing special devices. Some address only a single aspect of care. None of them combine AI mood detection with social features and health management.

2.1.8 Research Gap and Conclusion

The literature review showcases several important findings that support the development of the product, 'ElderConnect'

Growing Need: The elderly population is increasing rapidly, and social isolation is a serious problem affecting their health and well-being.

Technology Potential: The digital technology can help to addressing social isolation, but existing applications aren't designed with elderly users in mind.

Design Requirements: The research has identified specific design principles that make applications accessible and usable for elderly individuals.

AI Capabilities: The Artificial Intelligence, particularly sentiment analysis, can be effectively used to monitoring emotional well-being and provide support.

Market Gap: Existing solutions either focus on only one aspect of elderly care or require expensive proprietary hardware.

The research gap that ElderConnect addresses can be summarized as follow:

There is no comprehensive and accessible mobile application that combines social interaction, emotional well-being which monitored through AI, and health management in a single platform designed specifically for elderly users and their caretakers.

Existing platforms either:

- Are too complex for elderly users (Facebook, WhatsApp)
- Require special hardware that creates cost barriers (GrandPad)
- Focus on only one aspect of care (Medisafe, Connect2Affect)
- Lack of AI-powered emotional monitoring

ElderConnect addresses this gap by providing an all-in-one solution that is:

- Designed from the ground up for elderly accessibility
- Functional on standard smartphones without special hardware
- A combination of social, wellness, and health features in one platform
- Powered by AI for mood detection and personalization
- Connected to a caretaker dashboard for remote support

2.2 Market Research

2.2.1 Introduction

While the literature review provides the insights from existing academic research and available products, market research is essential to understand the specific needs and preferences of the target users. This section presents the findings from primary research conducted through surveys with elderly individuals and their caretakers.

The objectives of the market research were:

- To understand the current technology usage patterns among elderly users
- To identify challenges faced by elderly users with existing applications
- To assess the level of interest in the proposed ElderConnect features
- To gather insights from caretakers about their needs and concerns
- To validate the need for the proposed solution

2.2.2 Research Methodology

To gather primary data for this research, surveys were conducted with two target groups:

1. **Elderly Users:** Individuals of aged 55 and above who represent the primary users of the proposed application
2. **Caretakers:** Family members or caregivers who assist or monitor elderly individuals

Different data collection methods were used for each group based on their comfort with the technology. For the caretaker survey, an online questionnaire approach was selected as caretakers are generally comfortable with digital tools. This method allows reaching respondents conveniently and makes data collection and analysis easier.

For the elderly user survey, researcher used printed questionnaires instead of the online forms. This decision was made because many elderly individuals are not familiar with online forms and would struggle to complete them independently. By using printed media, it ensured that the respondents could participate the survey comfortably without technology barriers affecting their ability to provide genuine responses.

2.2.3 Survey Design

Two separate surveys were designed for the two target groups:

Survey 1: Caretaker Support Survey

This survey targeted family members and caregivers of elderly individuals. It consisted of 7 questions covering the relationship with the elderly person, frequency of assistance or monitoring, challenges faced in monitoring well-being and medication, interest in digital tools for monitoring, and willingness to use such tools.

Survey 2: Elderly User Survey

This survey targeted the individuals aged 55 and above. It consisted of 22 questions covering basic demographic information, technology usage patterns and confidence levels, difficulties faced with digital applications, social interaction and feelings of loneliness, interest in proposed features, and concerns about using such applications.

Both surveys used a mix of multiple-choice questions, checkbox questions (select all that apply), and open-ended questions for additional feedback. The complete questionnaires are included in Appendix A.

2.2.4 Data Collection

Responses were collected from:

- **52 elderly respondents** for the elderly user survey
- **13 caretaker respondents** for the caretaker survey

The caretaker survey was distributed online through social media and personal networks over a period of two weeks. The printed questionnaires were distributed through community centers, family networks, and local gatherings for the elderly users to conduct the survey. Family members or the researcher helped them to complete the survey for elderly respondents who needed assistance with reading or writing, while ensuring the responses reflected the elderly person's own views.

2.2.5 Caretaker Survey Results

This section presents the analysis of responses from the 13 caretakers who participated in the survey.

Question 2: Who do you currently live with?

Table 2: Showing living situations

Living Situation	Responses	Percentage
Alone	14	26.9%
With spouse/partner	18	34.6%
With children/family	15	28.8%
With a caretaker	3	5.8%
Other	2	3.8%
Total	**52**	**100%**

A notable 26.9% of respondents live alone, which puts them at higher risk of social isolation. Even those living with a spouse (34.6%) may experience isolation if both partners have limited social connections outside the home. This distribution highlights the importance of tools that can help elderly individuals stay connected with others.

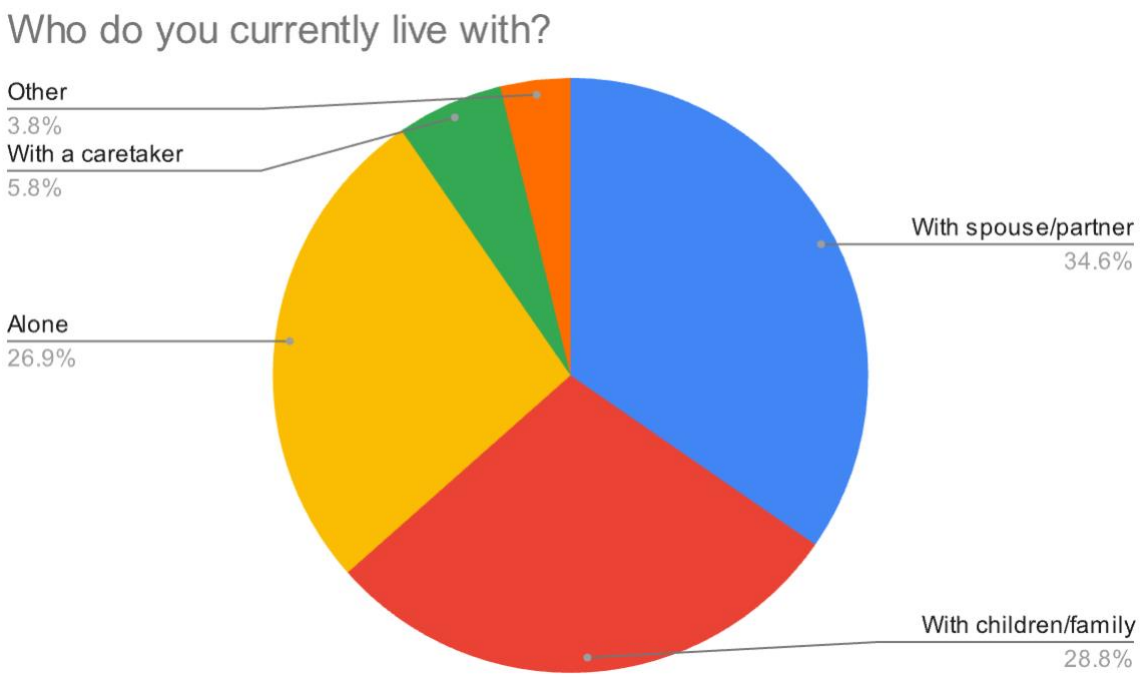

Figure 5: Pie chart showing living situations

Question 3: Do you find it difficult to monitor the elderly person's well-being or medication routine?

Table 3: Showing difficulty levels

Response	Responses	Percentage
Yes	5	38.5%
Sometimes	6	46.2%
No	2	15.4%
Total	**13**	**100%**

The results show that 84.6% of caretakers (combining "Yes" and "Sometimes") experience some level of difficulty in monitoring well-being or medication routines. Only 15.4% reported no difficulty at all. This strongly supports the need for tools that can assist with remote monitoring of elderly family members.

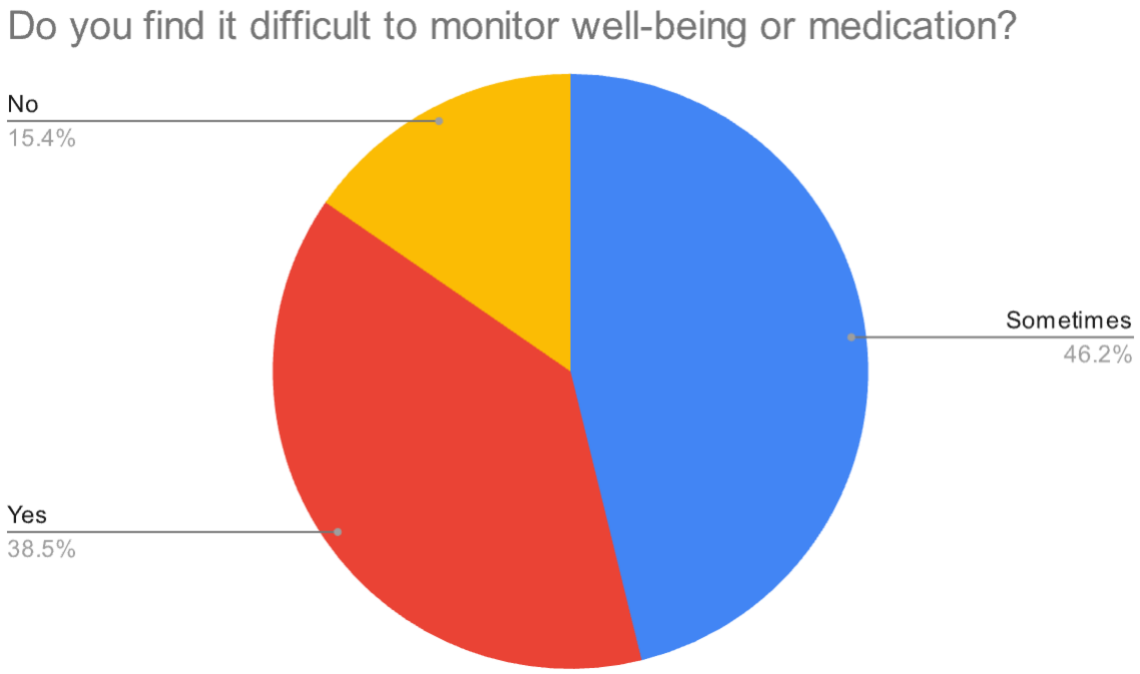

Figure 6: Pie chart showing difficulty levels

Question 4: Which challenges concern you the most? (Select all that apply)

Table 4: Showing challenges

Challenge	Responses	Percentage
Emergency situations	11	84.6%
Loneliness or mood changes	10	76.9%
Missed medication	9	69.2%
Difficulty using mobile apps	7	53.8%

Emergency situations were the highest concern among caretakers (84.6%), followed by loneliness or mood changes (76.9%). The high concern for mood changes validates the inclusion of AI mood detection in ElderConnect. Missed medication (69.2%) also ranked highly, supporting the medication reminder feature. The concern about elderly users having difficulty with mobile apps (53.8%) reinforces the need for an accessible, elderly-friendly interface.

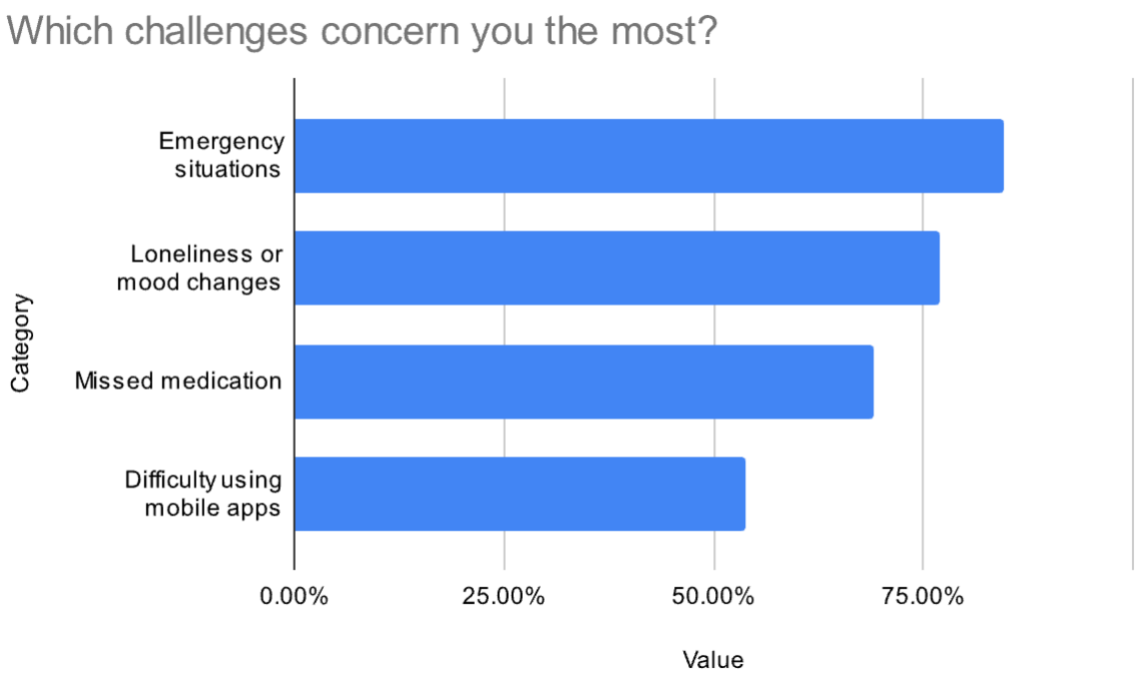

Figure 7: Horizontal bar chart showing challenges

Question 5: Would a simple digital dashboard to monitor mood and medication be helpful?

Table 5: Showing helpfulness rating

Response	Responses	Percentage
Very helpful	8	61.5%
Somewhat helpful	4	30.8%
Not helpful	1	7.7%
Total	**13**	**100%**

A significant majority (92.3%) found the idea of a digital dashboard either very helpful or somewhat helpful. Only one respondent indicated it would not be helpful. This provides strong validation for the caretaker dashboard feature planned in ElderConnect.

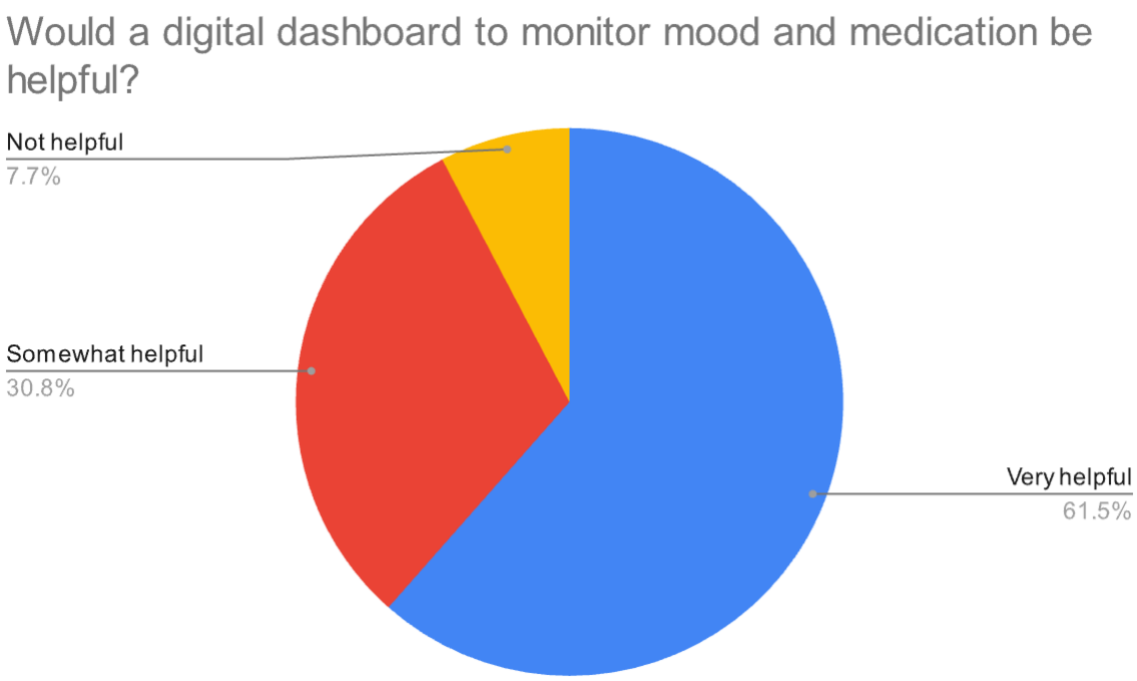

Figure 8: Pie chart showing helpfulness rating

Question 6: Would you be willing to use a simple mobile or web interface for this purpose?

Table 6: Showing willingness to use

Response	Responses	Percentage
Yes	9	69.2%
Maybe	3	23.1%
No	1	7.7%
Total	**13**	**100%**

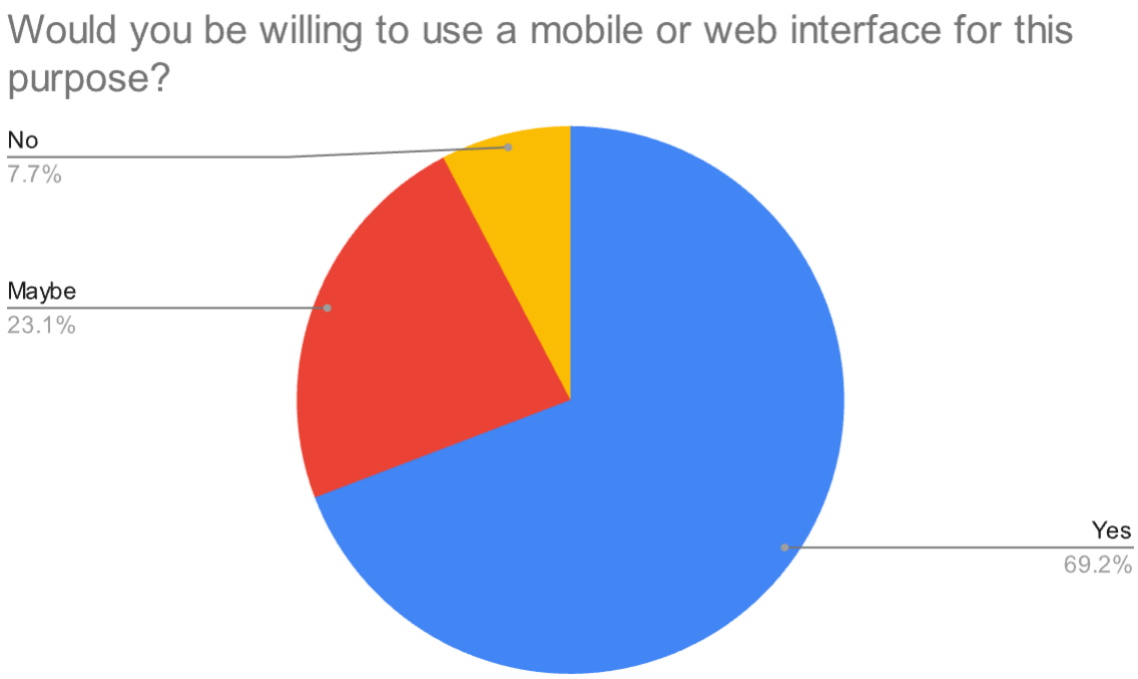

Figure 9: Pie chart showing willingness to use

The majority (69.2%) expressed definite willingness to use such an interface, with an additional 23.1% open to the possibility. Only one respondent was not interested. This indicates strong potential adoption among caretakers for the proposed caretaker portal.

2.2.6 Elderly User Survey Results

This section presents the analysis of responses from the 52 elderly respondents who participated in the survey.

Question 1: What is your age group?

Table 7: Showing age distribution

Age Group	Responses	Percentage
Below 55	4	7.7%
55-64	12	23.1%
65-74	24	46.2%
75+	12	23.1%
Total	**52**	**100%**

The majority of respondents (46.2%) were in the 65-74 age group, which represents the core target demographic for ElderConnect. When combined with the 75+ group (23.1%), nearly 70% of respondents were 65 or older. This age distribution ensures the survey findings are relevant to the intended user base.

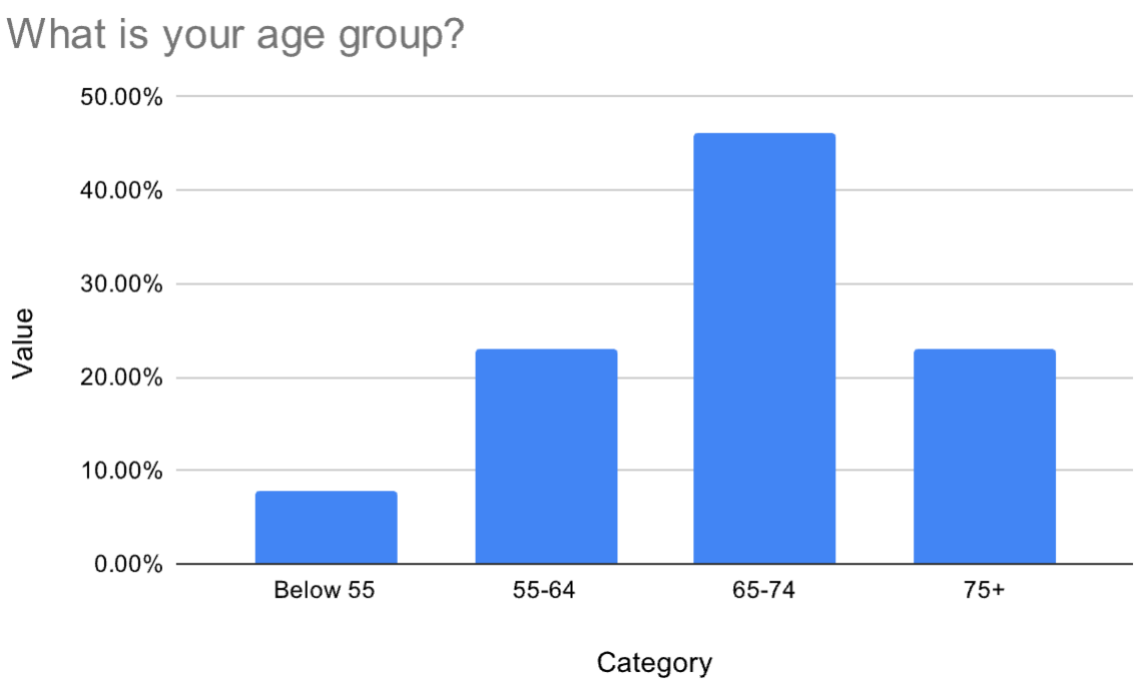

Figure 10: Bar chart showing age distribution

Question 3: How confident are you using a smartphone?

Table 8: Showing confidence levels

Confidence Level	Responses	Percentage
Very confident	5	9.6%
Confident	11	21.2%
Somewhat confident	24	46.2%
Not confident	12	23.1%
Total	**52**	**100%**

The majority of respondents (46.2%) described themselves as only "somewhat confident" with smartphones, and 23.1% were "not confident" at all. This means nearly 70% of respondents have limited confidence in using smartphones. These findings strongly support the need for an accessible, simple application design that does not assume technical expertise.

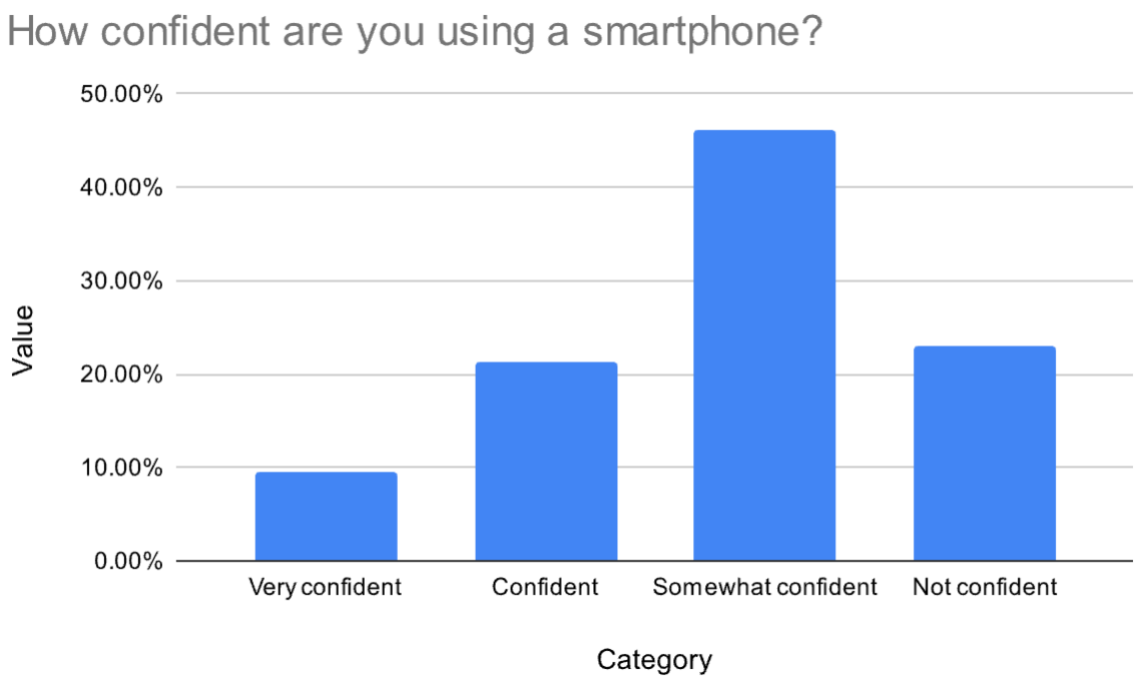

Figure 11: Bar chart showing confidence levels

Question 4: What difficulties do you face when using apps? (Select all that apply)

Table 9: Showing difficulties

Difficulty	Responses	Percentage
Small text / difficult to read	38	73.1%
Typing difficulties	33	63.5%
Complicated menus	31	59.6%
Fear of pressing wrong button	28	53.8%
Forgetting steps	25	48.1%
Lack of someone to teach me	19	36.5%

Small text was the most commonly reported difficulty (73.1%), followed by typing difficulties (63.5%) and complicated menus (59.6%). These findings directly inform the design of ElderConnect. The application will feature large text throughout, voice message options to reduce typing needs, and simplified navigation with minimal menu levels.

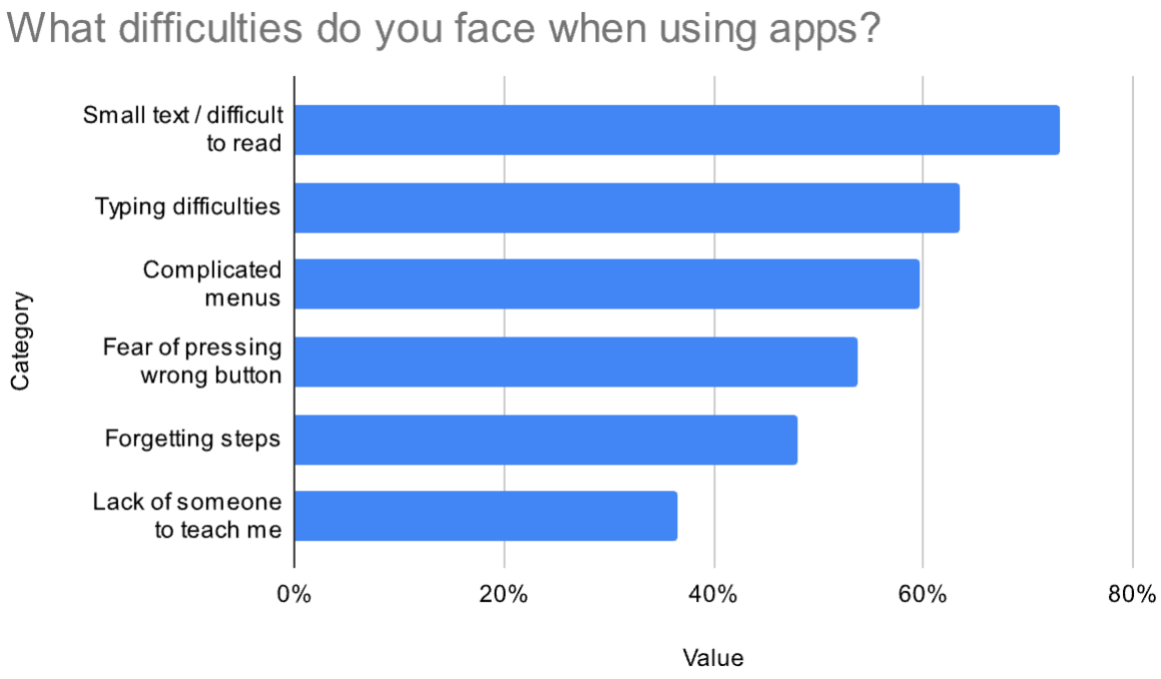

Figure 12: Horizontal bar chart showing difficulties

Question 5: Do you often feel lonely or disconnected?

Table 10: Showing loneliness levels

Response	Responses	Percentage
Yes	11	21.2%
Sometimes	27	51.9%
No	14	26.9%
Total	**52**	**100%**

A significant 73.1% of respondents experience loneliness either regularly (21.2%) or sometimes (51.9%). Only about a quarter of respondents (26.9%) reported not feeling lonely. This finding strongly supports the development of an application focused on social connection and emotional well-being.

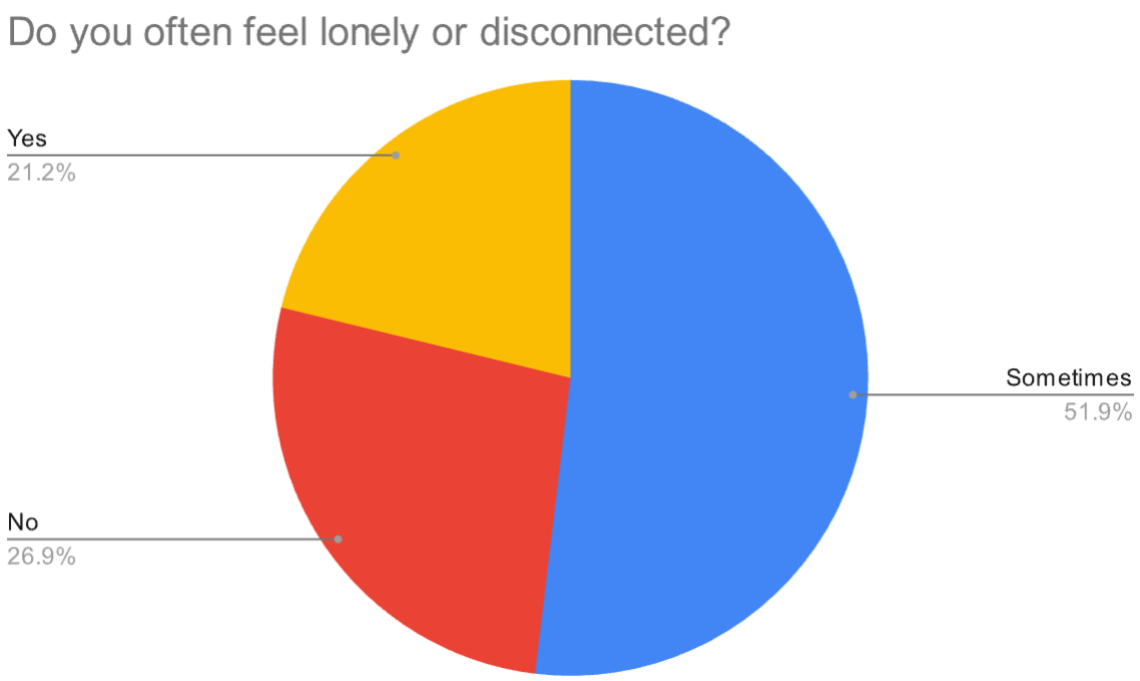

Figure 13: Pie chart showing loneliness levels

Question 6: Do you think a system that detects your emotional mood based on your messages would be useful?

Table 11: Showing interest in mood detection

Response	Responses	Percentage
Yes	22	42.3%
Maybe	21	40.4%
No	9	17.3%
Total	**52**	**100%**

The majority (82.7%) see potential value in mood detection, with 42.3% definitely finding it useful and 40.4% open to the idea. While some respondents (17.3%) were not interested, the overall positive response validates the AI sentiment analysis feature planned for ElderConnect.

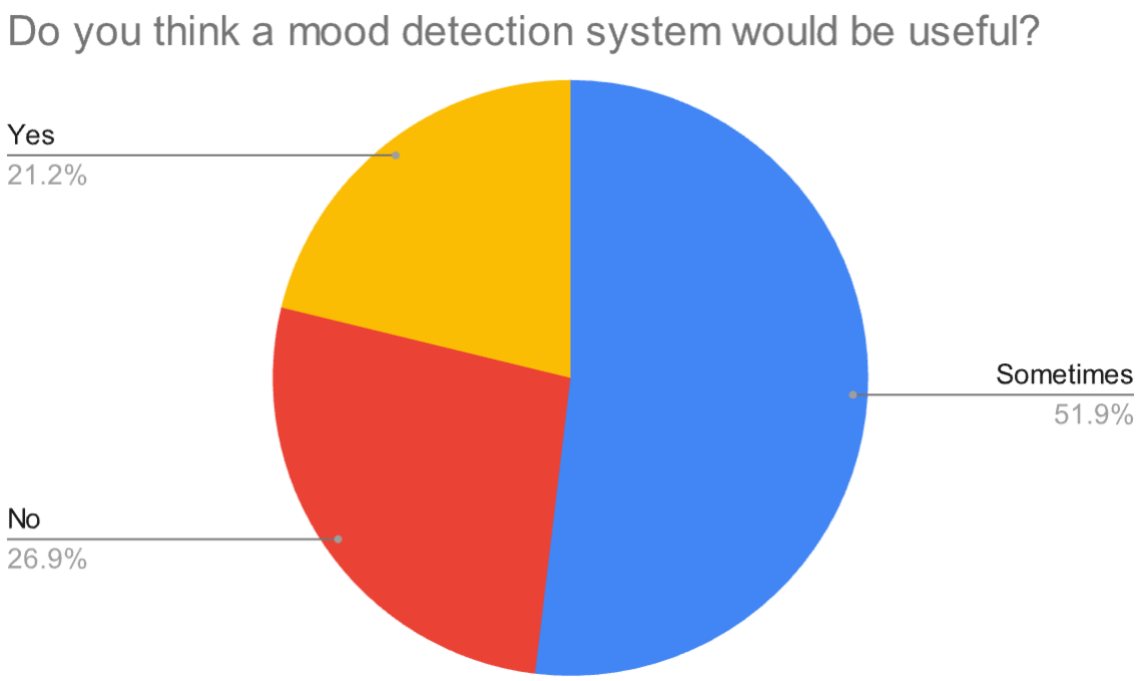

Figure 14: Pie chart showing interest in mood detection

Question 8: Would it be useful if the app reminded you when it's time to take your medicine?

Table 12: Showing interest in medication reminders

Response	Responses	Percentage
Yes	37	71.2%
Maybe	11	21.2%
No	4	7.7%
Total	**52**	**100%**

An overwhelming 71.2% of respondents would find medication reminders useful, with only 7.7% saying no. This strongly validates the medication reminder feature as a core component of ElderConnect.

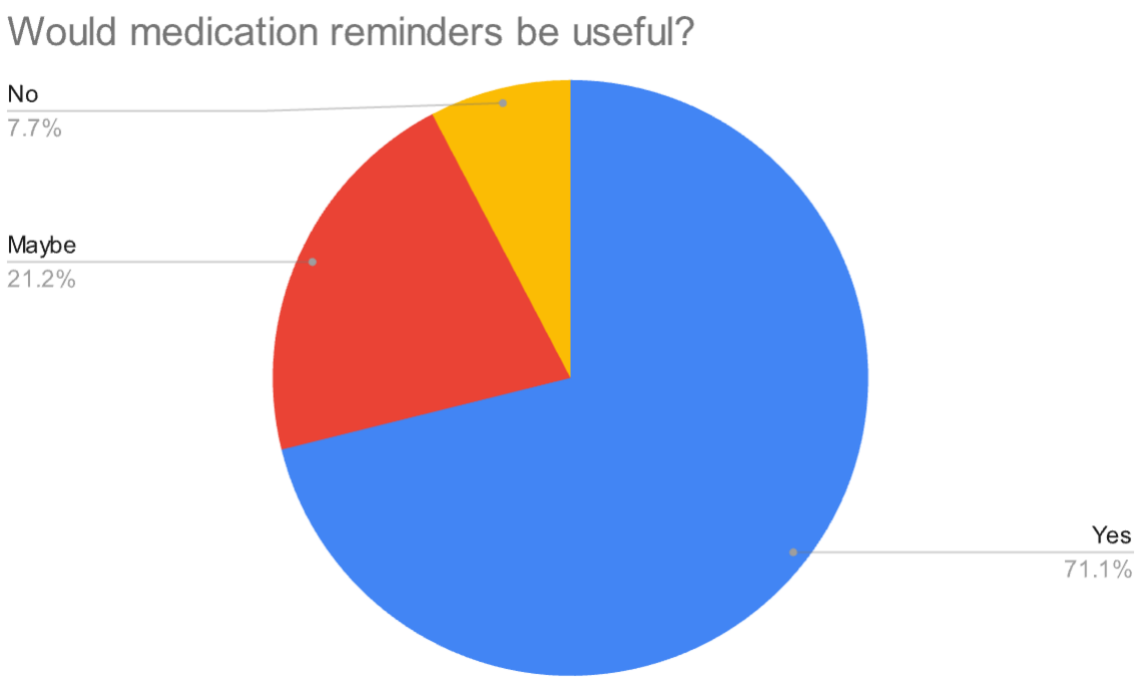

Figure 15: Pie chart showing interest in medication reminders

Question 10: Which features would be most helpful to you? (Select all that apply)

Table 13: Showing feature preferences

Feature	Responses	Percentage
Big buttons/clear text	44	84.6%
Emergency contact button	42	80.8%
Simple communication	40	76.9%
Medication reminders	38	73.1%
Voice instructions	32	61.5%
Personalized news	27	51.9%
Memory games	25	48.1%
Caretaker monitoring	24	46.2%
Mood detection	21	40.4%

The top three requested features were big buttons and clear text (84.6%), emergency contact button (80.8%), and simple communication (76.9%). Importantly, all features planned for ElderConnect received support from at least 40% of respondents, validating the proposed feature set.

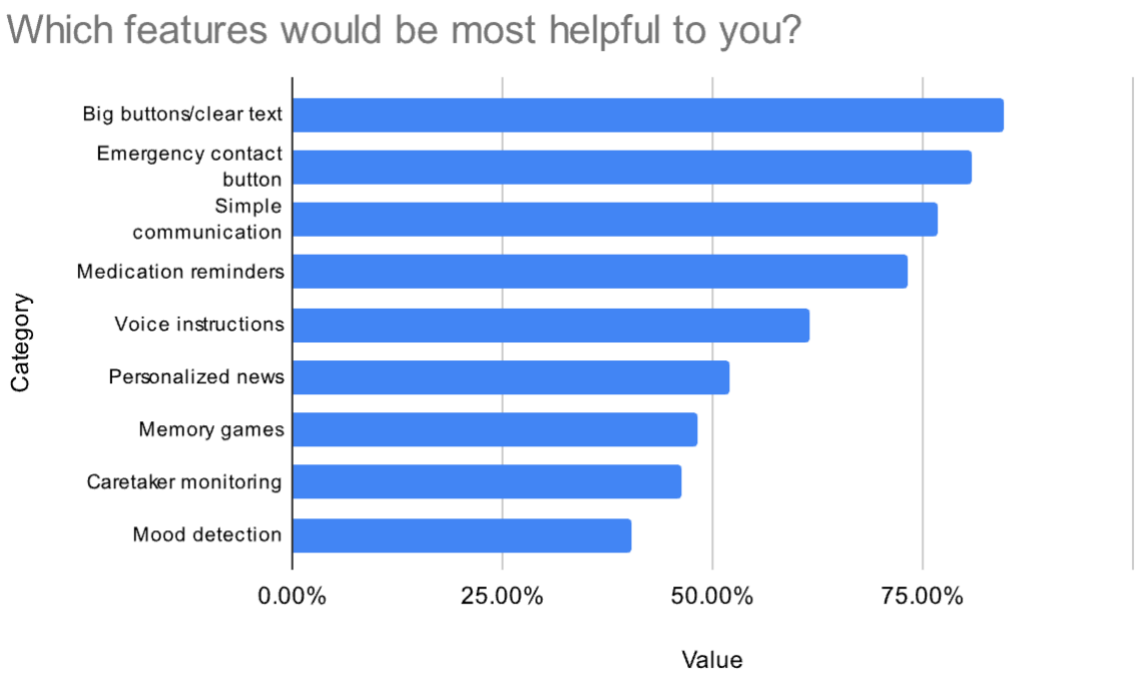

Figure 16: Horizontal bar chart showing feature preferences

Question 12: Was someone helping you complete this survey?

Table 14: Showing assistance needed

Response	Responses	Percentage
Yes	24	46.2%
No	28	53.8%
Total	**52**	**100%**

Nearly half (46.2%) of elderly respondents needed help completing the survey. This further demonstrates the accessibility challenges elderly users face with digital interfaces and reinforces the importance of designing ElderConnect to be as simple and intuitive as possible.

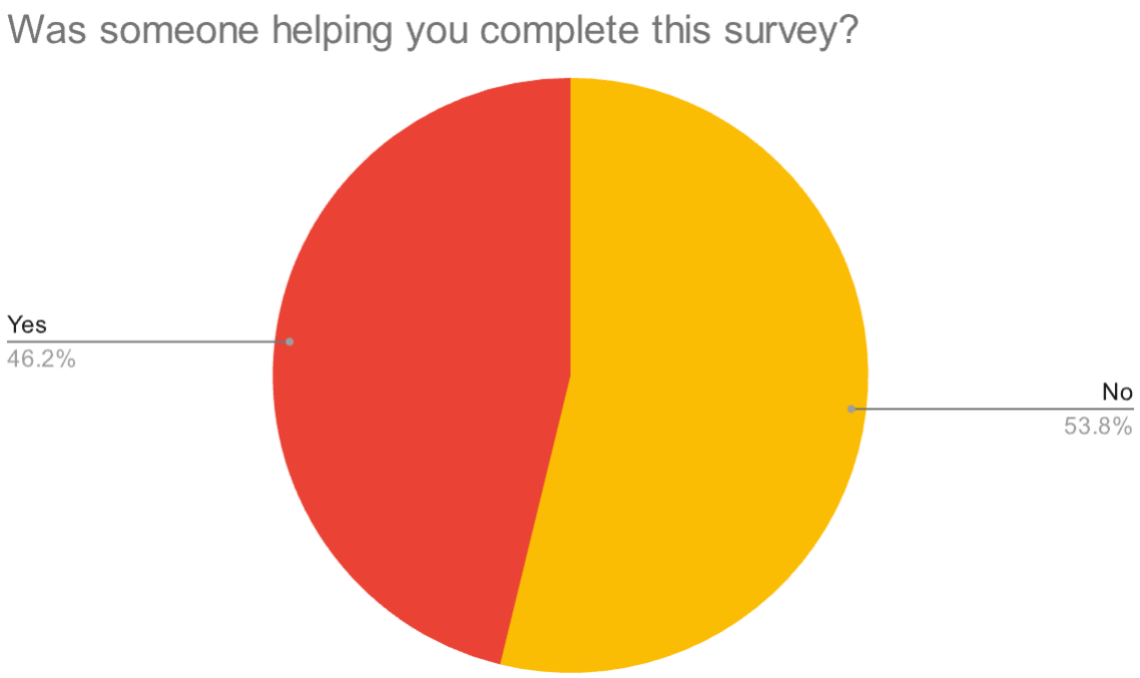

Figure 17: Pie chart showing assistance needed

2.2.7 Key Findings and Conclusion

The market research findings strongly support the development of ElderConnect. The key conclusions from both surveys are summarised below.

From Caretaker Survey:

- 84.6% of caretakers experience difficulty monitoring well-being and medication of elderly relatives
- Emergency situations (84.6%) and mood changes (76.9%) are the top concerns among caretakers
- 92.3% found the idea of a digital monitoring dashboard helpful
- 69.2% are definitely willing to use such a tool

From Elderly User Survey:

- The majority of respondents (69.3%) were aged 65 and above, representing the core target demographic73.1% face difficulties with small text, and 63.5% struggle with typing
- Nearly 70% of elderly respondents have limited confidence using smartphones65.4% forget medication at least sometimes
- 73.1% face difficulties with small text, and 63.5% struggle with typing82.7% see value in mood detection features
- 73.1% experience loneliness at least sometimes
- 82.7% see value in mood detection features
- 71.2% would find medication reminders useful
- 84.6% want big buttons and clear text as a priority feature
- 46.2% needed assistance to complete the survey, highlighting accessibility challenges

These findings validate the need for ElderConnect and confirm that the planned features address the real needs of both elderly users and their caretakers. The research also highlights the critical importance of accessibility and simplicity in the application design. Features must be easy to discover and use without requiring technical expertise or external help.

Project Plan

3.1 Project Overview

This chapter describes how the ElderConnect system will be planned for implementation. When developing a system, the project plan must be well structured according to the given timeframe. By using the Gantt chart and Work Breakdown Structure, this section explains how the proposed system will be planned and executed.

The project is based on the Agile approach that provides the opportunity to develop in iterations with the help of sprints. Every sprint brings with it certain features that can be tested and reviewed to proceed to the next one. This method is appropriate to ElderConnect as it enables the constant enhancement of it in response to feedback, which matters in the context of creating an application that the elderly population can use and has their own accessibility requirements.

The ElderConnect project began in October 2025 and is planned to be executed by April 2026, which leaves the whole lifecycle of a project at about six months. The project is broken down into seven key stages which are Planning, Requirement Gathering and Research, Design, Development (six sprints), Testing, Evaluation and Documentation.

3.2 Gantt Chart

The Gantt chart is a type of bar chart that illustrates the project schedule. It shows when each task is expected to start and finish, and how different phases relate to each other over time. The Gantt chart helps to track progress and stay on schedule throughout the project.

The development phase is divided into six sprints following the Agile methodology:

- **Sprint 1:** Setup and Authentication - Flutter environment setup, MongoDB integration, voice-guided registration and login
- **Sprint 2:** Social Features - Social feed, post creation, photo upload, Talk Button for voice messages
- **Sprint 3:** AI Integration - Hugging Face API connection, sentiment analysis, mood scoring algorithm
- **Sprint 4:** Health and Wellness - Medication reminders, wellness games, breathing exercises, emergency contact
- **Sprint 5:** Caretaker Portal - Dashboard interface, medication management, mood monitoring, data sync
- **Sprint 6:** Accessibility and Refinement - Text-to-speech, colour-coded navigation, bug fixes

Each sprint includes testing and review at the end to make sure the features work correctly before moving forward.

The Gantt chart for ElderConnect is presented in Figure 20 below.

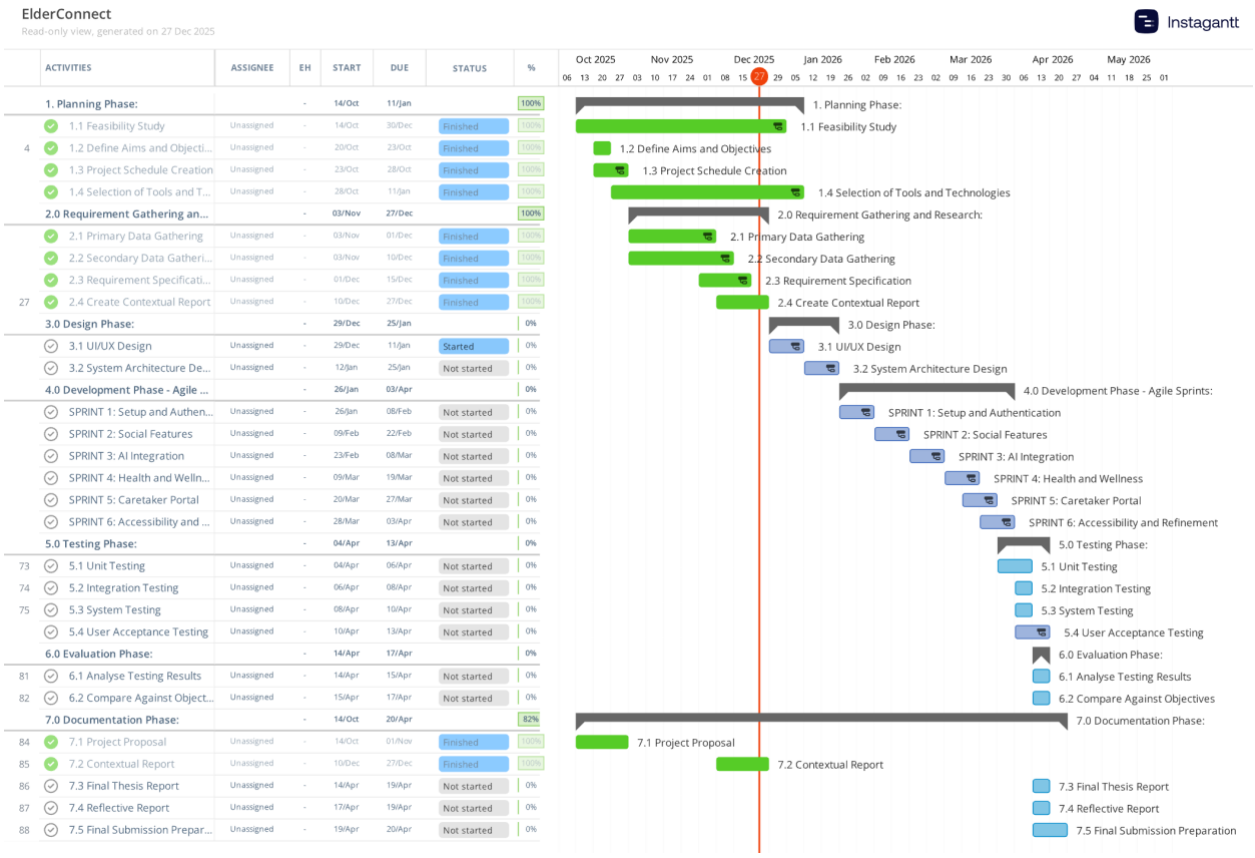

Figure 18: Gantt Chart for ElderConnect project

3.3 Work Breakdown Structure

The Work Breakdown Structure (WBS) is a chart that divides the entire workload into smaller, manageable parts. Dividing the project into these sections will allow seeing what should be done at each step and make it easier to estimate the time that should be spent on completing each task. It is also easier to monitor progress in the project using the WBS.

The WBS for ElderConnect is organized into seven main phases:

1.0 Planning - This phase covers feasibility study, defining aims and objectives, creating the project schedule (Gantt chart and WBS), and selecting the tools and technologies (Flutter, MongoDB, Hugging Face API).

2.0 Requirement Gathering and Research - This phase includes primary data gathering through surveys with elderly users and caretakers, secondary data gathering through literature review and market analysis, requirement specification with user stories and functional requirements, and creating the contextual report.

3.0 Design - This phase covers UI/UX design with accessibility focus (wireframes with large buttons and high contrast, high-fidelity prototypes), and system architecture design (database schema, AI logic flow, UML diagrams).

4.0 Development - This phase is divided into six sprints as described above. Each sprint focuses on specific features and includes its own testing and review cycle.

5.0 Testing - This phase includes unit testing, integration testing, system testing, and user acceptance testing with actual elderly users to check accessibility.

6.0 Evaluation - This phase involves analyzing testing results and comparing the final product against the original objectives.

7.0 Documentation - This phase runs throughout the project and includes the project proposal, contextual report, final thesis report, and reflective report.

The Work Breakdown Structure for ElderConnect is presented in Figure 19 below.

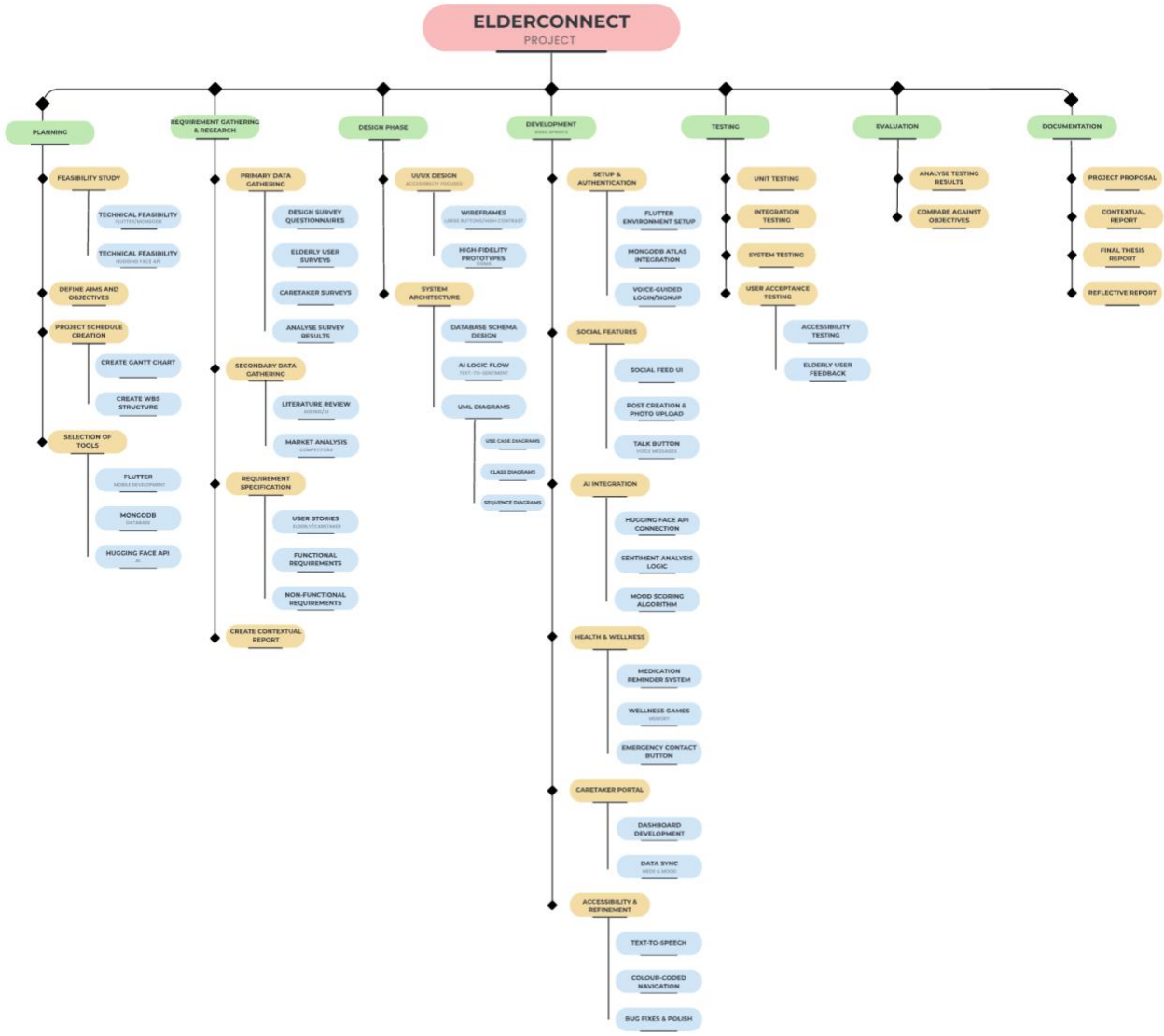

Figure 19: Work Breakdown Structure for ElderConnect Project

3.4 Current Progress

As of the contextual report submission date (27th December 2025), the following progress has been made on the project:

Completed:

- Phase 1.0 (Planning) has been fully completed, including feasibility study, aims and objectives, project schedule, and tool selection.
- Phase 2.0 (Requirement Gathering and Research) has been fully completed, including surveys with 52 elderly users and 13 caretakers, literature review, and market analysis.
- The contextual report has been completed and submitted.

In Progress:

- Phase 3.0 (Design) has started, with approximately 50% of the work completed.

Remaining:

- Complete the remaining design tasks
- Complete all six development sprints
- Complete testing and evaluation
- Complete final documentation and submission

The project is currently on schedule according to the planned timeline. No major delays or issues have been faced so far.

Planning the Artefact

4.1 Methodology

The methodology is a list of guidelines that are employed to come up with a system in a systematic way. Software development methodologies are not few but each one has its benefits and drawbacks. When choosing the methodology, it is necessary to keep in mind which approach would be best to use on the system under development.

In the case of ElderConnect project, the Agile approach has been chosen. Agile is an iterative method that breaks down the development process into smaller processes known as sprints. A working piece of the system is delivered by each is then tested and reviewed before proceeding to next sprint.

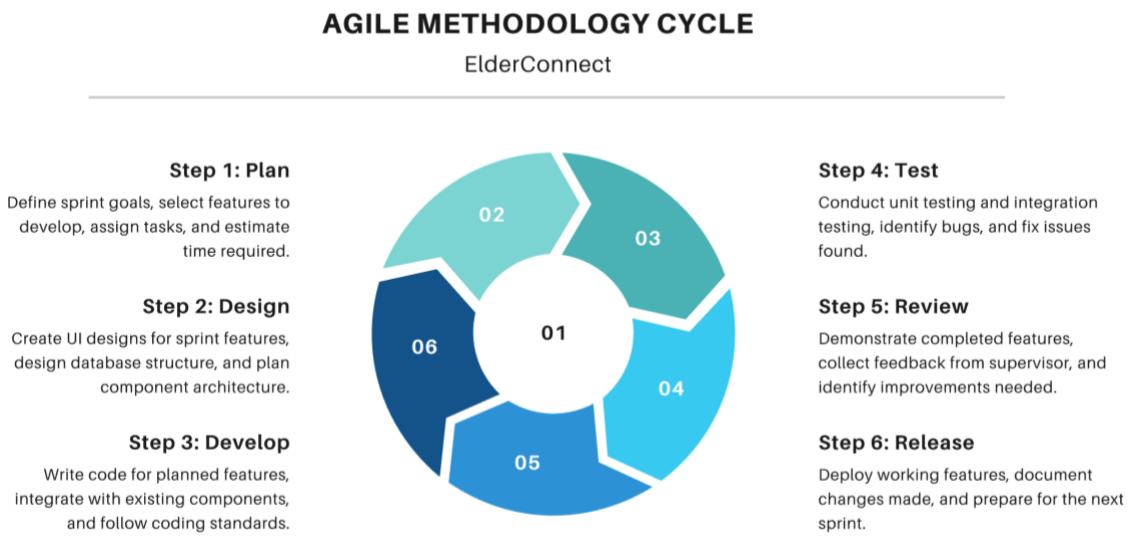

Figure 20: Agile method for ElderConnect

The Agile methodology was chosen for several reasons:

- **Flexibility:** Agile allows changes to be made during development based on feedback. This is important for ElderConnect because elderly users have specific accessibility needs that may require adjustments as testing progresses.
- **Continuous Testing:** Each sprint includes its own testing phase, which helps to identify and fix issues early rather than waiting until the end of the project.
- **User Feedback:** Agile encourages regular feedback from users. For this project, getting feedback from elderly users throughout development will help ensure the application meets their needs.

- **Manageable Workload:** Breaking the project into six sprints makes the workload easier to manage within the given timeframe.

4.2 Requirement Gathering and Analysis

The requirements for ElderConnect were gathered using both primary and secondary data collection methods.

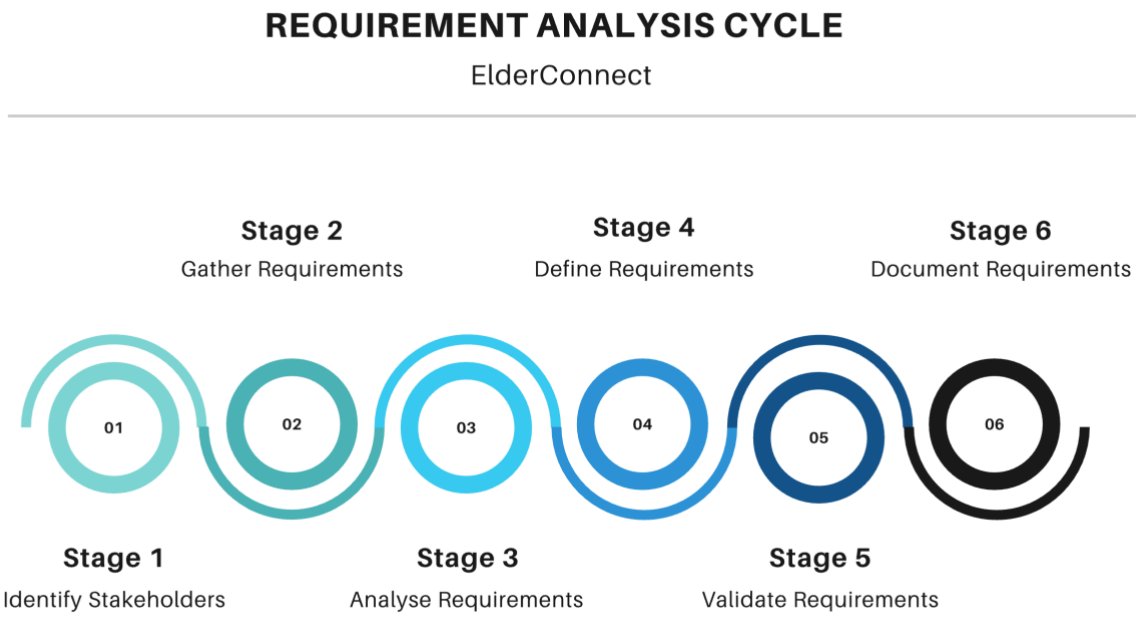

Figure 21: Requirement Analysis for ElderConnect

Primary Data Gathering:

Two surveys were conducted to understand the needs of the target users:

- The survey consisted of 52 senior users who were interviewed using print questionnaires to gain information on their use, the challenges they encounter when using the current apps, their loneliness, and the interest they show in new features.
- A survey with 13 caretakers using an online form to understand their challenges in monitoring elderly relatives and their interest in a caretaker dashboard.

The survey findings confirmed that elderly users face significant difficulties with existing applications and would benefit from features like large buttons, voice messages, medication reminders, and mood detection.

Secondary Data Gathering:

Academic research papers, journal articles, and existing systems were reviewed to understand the current state of technology for elderly care. This included reviewing similar applications like GrandPad, Connect2Affect, and Medisafe to identify gaps that ElderConnect can address.

Requirements Analysis:

Based on the gathered data, the following key requirements were identified:

Functional Requirements:

- User registration with voice-guided instructions
- Social feed for sharing posts and photos
- Talk Button for voice messages
- AI-based mood detection from text
- Medication reminder system with notifications
- Wellness section with memory games
- Caretaker dashboard for monitoring

Non-Functional Requirements:

- Simple and accessible interface with large buttons
- High contrast colours for visibility
- Fast response time
- Secure data storage
- Works on Android and iOS devices

4.3 Design

The design phase focuses on creating the visual and technical structure of the application. The design follows accessibility guidelines to ensure elderly users can navigate the application comfortably.

User Interface Design:

The interface is designed with the following principles:

- Large buttons (minimum 44x44 pixels)
- High contrast colours
- Simple navigation with minimal levels
- Colour-coded sections for easy recognition
- Clear text with large font sizes
- Voice guidance where needed

Wireframes and high-fidelity prototypes are created using Figma to visualise the interface before development begins.

System Architecture:

The application uses a client-server architecture with the following components:

- **Frontend:** Flutter framework for cross-platform mobile development (Android and iOS)
- **Backend:** MongoDB for database storage
- **Authentication:** MongoDB Atlas App Services for user login and session management
- **AI Integration:** Hugging Face API for sentiment analysis and mood detection

Database Design:

The MongoDB database stores the following collections:

- Users (profile information, interests, settings)
- Posts (social feed content, photos)
- Medications (reminder schedules, pill details)
- Mood Logs (sentiment analysis results)
- Caretaker Links (connections between elderly users and caretakers)

UML Diagrams:

Figure 22 presents the Use Case diagram showing how different users interact with the system.

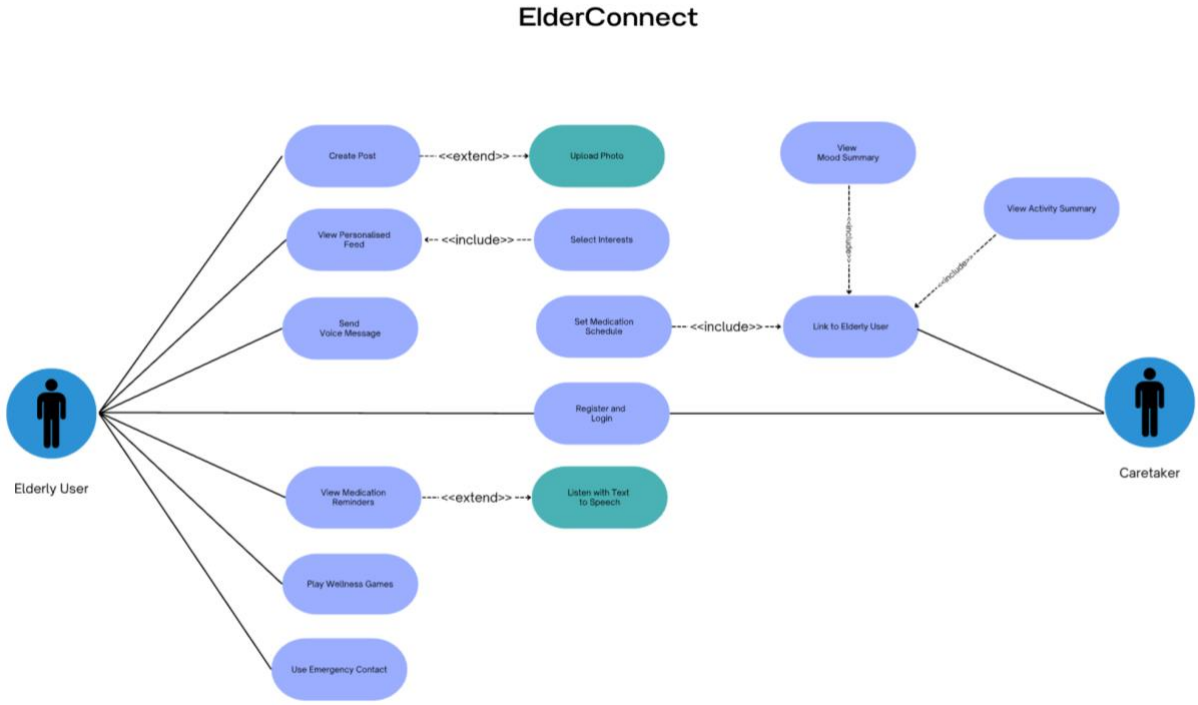

Figure 22: Use Case Diagram for ElderConnect

The Use Case diagram shows two main actors:

- **Elderly User:** Can register, login, create posts, send voice messages, view personalised content, play wellness games, receive medication reminders, and use emergency contact.
- **Caretaker:** Can register, login, set medication schedules, view mood summaries, monitor activity, and receive alerts.

4.4 Implementation

The implementation phase involves developing the application according to the design specifications. As mentioned earlier, the development follows six sprints:

Sprint 1 - Setup and Authentication:
Setting up the Flutter development environment, connecting MongoDB Atlas, and implementing voice-guided registration and login.

Sprint 2 - Social Features:
Developing the social feed interface, post creation with photo upload, and the Talk Button for voice messages.

Sprint 3 - AI Integration:
Connecting the Hugging Face API, implementing sentiment analysis logic, and developing the mood scoring algorithm.

Sprint 4 - Health and Wellness:
Building the medication reminder system with notifications, wellness games including memory exercises, and the emergency contact button.

Sprint 5 - Caretaker Portal:
Developing the caretaker dashboard interface, medication management features, mood monitoring display, and data synchronization.

Sprint 6 - Accessibility and Refinement:
Implementing text-to-speech functionality, finalizing colour-coded navigation, adjusting font scaling, and fixing bugs.

Tools and Technologies:

Table 15: Tools and Technologies

Component	Technology
Mobile App Development	Flutter (Dart)
Database	MongoDB
Authentication	MongoDB Atlas App Services
AI Sentiment Analysis	Hugging Face API
IDE	Visual Studio Code, Android Studio
Version Control	GitHub
UI Design	Figma
Testing Devices	Android and iOS smartphones

4.5 Testing Strategy

Testing is essential to ensure the application works correctly and meets user expectations. The following testing methods will be used

Unit Testing:
Testing individual functions and components to ensure they work as expected. This includes testing the login function, post creation, medication reminders, and mood detection.

Integration Testing:
Testing how different components work together. This includes testing the connection between the app and MongoDB, the AI integration with Hugging Face API, and data sync between elderly app and caretaker portal.

System Testing:
Testing the complete application to ensure all features work together as a whole system.

User Acceptance Testing (UAT):
Testing with actual elderly users to check if the application meets their needs. This includes:

- Accessibility testing (screen readers, font scaling, colour contrast)
- Usability testing to see if elderly users can navigate the app without help
- Collecting feedback on what works well and what needs improvement

4.6 Evaluation Strategy

After testing is completed, the application will be evaluated against the original objectives to determine how well the project goals have been achieved.

The evaluation will include:

- Comparing each objective with the delivered features
- Analyzing feedback from elderly users and caretakers
- Identifying what worked well and what could be improved
- Documenting lessons learned for future projects

4.7 Risks and Mitigation

Every project has potential risks that need to be considered. The table below shows the identified risks and how they will be handled:

Table 16: Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Difficulty integrating AI sentiment model	Medium	High	Use pre-trained Hugging Face API; have backup rule-based approach
Challenges in designing accessible interface	High	High	Conduct usability testing with elderly users; refine based on feedback
Data privacy issues	Medium	High	Implement secure authentication; encrypt sensitive data; consent-based sharing
Data handling errors during development	Medium	Medium	Use structured validation; frequent backups; thorough testing
Time constraints for completing all features	Medium	Medium	Prioritize core features; use sprint reviews to adjust scope if needed

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Appendices

Appendix A: Elderly User Survey Questionnaire

Survey on Technology Use, Social Interaction, and Health Support Among Older Adults

Consent

1. Do you agree to participate in this survey?
    - Yes, I agree
    - No, I do not agree

Basic Information

2. What is your age group?
    - Below 55
    - 55-64
    - 65-74
    - 75+
3. Gender (optional)
    - Male
    - Female
    - Prefer not to say
4. Who do you currently live with?
    - Alone
    - With spouse/partner
    - With children/family
    - With a caretaker
    - Other

Technology Usage

5. Do you use a smartphone regularly? If YES, how confident are you using a smartphone?
    - Very confident
    - Confident
    - Somewhat confident

- Not confident

6. How often do you use the following? (Phone calls, Text messaging, Video calls, Social media, News apps/websites)
    - Daily
    - Weekly
    - Monthly
    - Rarely
    - Never
7. Do you face difficulties using digital apps?
    - Yes
    - No
    - Sometimes
8. (If Yes/Sometimes) What difficulties do you face? (Tick all that apply)
    - Small text / difficult to read
    - Complicated menus
    - Forgetting steps
    - Typing difficulties
    - Fear of pressing the wrong button
    - Lack of someone to teach me

Social Interaction and Loneliness

9. How often do you communicate with friends or family online?
    - Daily
    - A few times a week
    - A few times a month
    - Rarely
    - Never
10. Do you feel socially connected to your family and friends?
    - Yes
    - Somewhat

- No

11. Do you often feel lonely or disconnected?
    - Yes
    - Sometimes
    - No

Emotional Well-being

12. How often do you share your thoughts or feelings through text messages or posts?
    - Often
    - Sometimes
    - Rarely
    - Never
13. Do you think a system that detects your emotional mood based on your messages would be useful?
    - Yes
    - Maybe
    - No

Health and Medication Management

14. Do you take daily medications? If yes, have you ever forgotten your medication?
    - Often
    - Sometimes
    - Rarely
    - Never
15. Would it be useful if the app reminded you when it is time to take your medicine?
    - Yes
    - Maybe
    - No
16. Would you like a trusted caretaker to help manage your reminders through their device?
    - Yes
    - Maybe

- No

Preferred Features

17. Which features would be most helpful to you? (Tick all that apply)
    - Simple communication
    - Personalised news
    - Big buttons/clear text
    - Voice instructions
    - Mood detection
    - Memory games
    - Medication reminders
    - Emergency contact button
    - Caretaker monitoring
18. What concerns would you have about using such an app?
    - Privacy concerns
    - Too difficult to learn
    - Too many features
    - Not needed
19. Any additional suggestions to make an app more suitable for older adults? (Open text response)

Submission

20. Was someone helping you complete this survey?
    - Yes
    - No

Appendix B: Caretaker Survey Questionnaire

Caretaker Support Survey – Elderly Care

1. What is your relationship with the elderly person?
    - Son/Daughter
    - Spouse

- Relative
- Caretaker

2. How often do you assist or monitor the elderly person?
    - Daily
    - Weekly
    - Occasionally
3. Do you find it difficult to monitor the elderly person's well-being or medication routine?
    - Yes
    - Sometimes
    - No
4. Which challenges concern you the most? (Select all that apply)
    - Missed medication
    - Loneliness or mood changes
    - Difficulty using mobile apps
    - Emergency situations
5. Would a simple digital dashboard to monitor mood and medication be helpful?
    - Very helpful
    - Somewhat helpful
    - Not helpful
6. Would you be willing to use a simple mobile or web interface for this purpose?
    - Yes
    - Maybe
    - No
7. Any additional suggestions? (Optional)
   (Open text response)